

FYP Proposal Submission Form

Detail of Project Members:	MUHAMMAD BILAL ZOHAIK KHAN
Supervised By:	Dr. Muhammad Sharif
Title:	Vendra: A Multi-Vendor Digital Marketplace for Global Commerce
Which real world Problem shall be solved by this project?	In Pakistan, online delivery and marketplace platforms frequently face order failures due to inaccurate stock availability, limited vendor digitization, weak trust between buyers and sellers, and lack of transparent dispute handling. Customers often place orders that get cancelled because vendors do not maintain real-time inventory. Vendors also lack affordable POS tools to manage stock, pricing, and online orders together. Riders face uncertain compensation rules and delays in payouts, which impacts service quality. This FYP proposes a scalable marketplace that integrates a vendor POS with dual inventory (private actual stock and public visible stock), escrow-based payments to reduce fraud and disputes, and a policy-driven order lifecycle for customers, vendors, and riders. The solution improves stock accuracy, increases trust, reduces cancellations, and enables a sustainable business model with fair rider payments.

Project Description (What, How and Objectives):

Overall working/summary of this FYP:

The proposed system is a multi-vendor e-commerce and delivery platform that connects customers, vendors, riders, and an organization/admin through secure role-based access. Vendors operate a POS-integrated inventory system that maintains private stock (actual quantity) and public stock (customer-visible quantity controlled by policy rules). Customers browse products, place orders, and pay through an escrow mechanism where funds are held until delivery is completed or disputes are resolved. When an online order arrives, the vendor POS creates a reservation to avoid overselling, staff picks and packs the order, and marks it ready for pickup. Riders receive delivery tasks, update delivery states (picked, on the way, delivered), and earn through a payout formula based on base fee, distance, and waiting time rules. The admin module approves vendors/riders, manages categories, defines policy rules (cancellations, refunds, penalties, visibility limits), and monitors platform operations. The platform earns through configurable commission and service fees, and can later extend to subscriptions and promotions. This solution targets improved inventory accuracy, reduced order failures, stronger trust, and scalable deployment for Pakistan with potential international expansion.

Name and detail of each module in FYP:

1. Authentication & RBAC Module: Signup/login, JWT sessions, and role-based access for Admin, Vendor, Rider, Customer. 2. Customer Module: Browse stores/products, cart, checkout, order tracking, reviews, complaint/refund requests. 3. Vendor Module: Store onboarding, product catalog, pricing, staff roles (cashier/inventory/packer), order handling. 4. POS & Inventory Module: Stock in/out, stock ledger, walk-in sales, online order reservation, pick/pack workflow, dual inventory (private/public) calculation. 5. Order Management Module: End-to-end order lifecycle states and history. 6. Rider & Delivery Module: Task assignment, delivery status updates, delivery proof, rider availability. 7. Escrow & Payment Module: Hold funds, release on delivery, refund rules, payout settlement logic. 8. Policy & Dispute Module: Rules for cancellations, penalties, refund decisions, evidence (photos/timestamps), dispute resolution. 9. Admin/Organization Module: Vendor/rider approvals, policy configuration, commission/service fee settings, analytics dashboard. 10. Notifications Module: Email/SMS/in-app alerts for order and delivery updates.

Member-wise Module Information:

1: Module developed by MUHAMMAD BILAL (CIIT/SP23-BCS-045/WAH)?

Will develop the Vendor, POS and Inventory Management Module which includes barcode scanning. This includes vendor onboarding, product and pricing management, stock in/out with stock ledger, walk-in POS sales, online order reservation and packing workflow, and private-to-public inventory visibility rules. Member 1 will also implement admin controls related to vendor approval and catalog/category permissions. When an order is accepted, the Items in the order are removed from the public inventory automatically so the vendor does not have to scan the items again.

2: Module developed by ZOHAIB KHAN (CIIT/SP23-BCS-088/WAH)?

Member 2 will develop the Customer, Rider, Order, and Escrow/Dispute Modules. This includes customer browsing and checkout, order lifecycle management, escrow hold/release/refund flow, rider onboarding and delivery tracking, rider payout calculation and earnings dashboard, and dispute handling with evidence and policy-based decisions. Member 2 will also implement admin workflows for rider approval and dispute resolution. Rider module will not be integrated with BYKEA, it will be our own.

Similar FYP from RMS student console

1. Multi-Vendor Product for shopping malls 2. Multivendor eCommerce Marketplace (Wah Mart) 3. shoppiee.com (a multivendor E-commerce website)

Three (3) new, but main, functionalities adding to this FYP:

Escrow-based payment handling, where customer payments are securely held by the platform and released only after successful order delivery or dispute resolution, unlike direct vendor payments used in

2. Policy-driven order lifecycle management that defines structured rules for order acceptance, cancellation, refunds, penalties, and dispute resolution among customers, vendors, and riders.

	3. Integrated delivery workflow with rider management, including order assignment, delivery tracking, and controlled payout calculation, which was not implemented in earlier multivendor marketplace FY
Project Streams:	